Project 4: TCGA RNA-seq data analysis

The goal of this project is to streamline the process of visualizing expression profile of a gene of interest for a particular solid tumor. For example, we are working with Lung Adenocarcinoma (LUAD), a type of non-small cell lung cancer (see the image below).

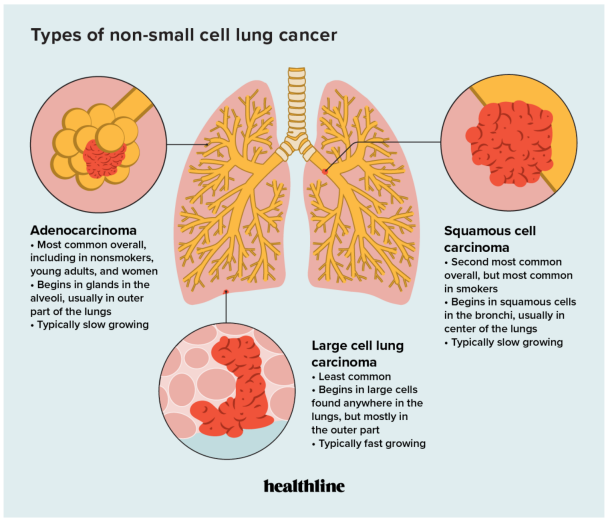
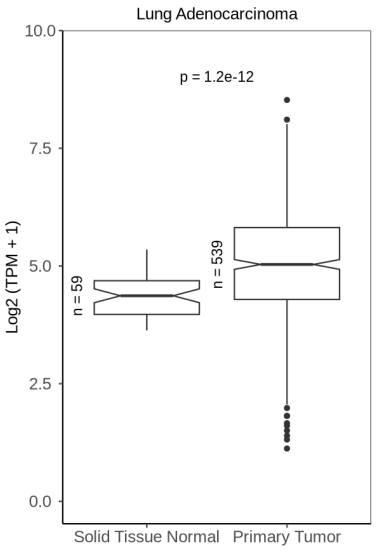


Illustration by Jason Hoffman

https://tinyurl.com/5b3v8bzk

In cancer field, researchers attempt to find the biomarkers i.e., certain protein of other macromolecules which can tell presence of tumor in a precise manner. In this project, your goal is to quantify if existing biomarkers have significantly different expression profile in tumor tissue compared to the adjacent healthy tissue. I have shared the data relevant to this. Here are the details of the data:



At this URL, you will find the following datasets:

https://figshare.com/s/fd7276e3583b457bd61d

1. Sample sheet file (gdc_sample_sheet.tsv)
2. Manifest file (gdc_manifest.txt)
3. Gene expression data (tcga_data.tar.gz)

You should run the command: tar -zxvf tcga_data.tar.gz to unzip and get the real data file.
The sample sheet file has tissue biopsy information, like it is “Primary Tumor” or it is “Solid Tissue Normal”. There might be some other categories, but we are not interested in that for this project. You goal is to generate a gene expression boxplot for “**NKX2-1**” gene in Primary Tumor (PT) and Solid Tissue Normal (STN) conditions. See example boxplot in right. X-axis is the two conditions, and Y-axis is the Log2 of expression value + 1 (TPM + 1).

HINT:
e6e83fd4-2f80-44e1-8dce-588a37308dbb is one example of Solid Tumor Normal in gdc_sample_sheet.tsv. So, for this there is a corresponding directory in the gene expression data file. It will be “e6e83fd4-2f80-44e1-8dce-588a37308dbb/f4a76e18-60fa-4ef2-91ef-e2d7a1d11c8d.rna_seq.augmented_star_gene_counts.tsv”

You should search for NKX2-1 in this file and get the 7th column value (tpm_unstranded). The value is: “53.1718”.